

DECENT COACHING CLASSES

X CBSE

Academic Conquest - II — ANSWER KEY

SECTION A — MCQ Answers

1. (a) gets oxidised — Zinc loses electrons (displaces H₂ from the acid), so it is oxidised. [1]
2. (c) strongly acidic — A pH value less than 7, closer to 0, indicates a strongly acidic solution. [1]
3. (c) carbon dioxide — $\text{NaHCO}_3(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ [1]
4. (b) decomposition reaction — $2\text{AgCl}(\text{s}) \xrightarrow{\text{Sunlight}} 2\text{Ag}(\text{s}) + \text{Cl}_2(\text{g})$ [1]
5. (b) Sodium carbonate — It is a salt of a strong base (NaOH) and a weak acid (H₂CO₃), so its solution is basic (pH > 7). [1]

SECTION B — Answers

6. Balanced equation: $2\text{KOH} + \text{Al}_2(\text{SO}_4)_3 \rightarrow 3\text{K}_2\text{SO}_4 + 2\text{Al}(\text{OH})_3$
(Check: K = 2 on both sides after multiplying KOH by 2 and K₂SO₄ by 3; Al = 2 on both sides; SO₄ = 3 on both sides; OH = 6 on both sides.) [2]
7. A magnesium ribbon is normally coated with a thin layer of magnesium oxide on its surface (formed by reaction with atmospheric oxygen/moisture). This layer prevents the ribbon from burning properly. Cleaning it with sandpaper removes this oxide layer and exposes the fresh metal surface, allowing it to burn easily in air. [2]
8. Exothermic reactions release heat along with the formation of products, e.g., burning of natural gas: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g}) + \text{heat}$. Endothermic reactions absorb energy (heat/light/electricity) during the reaction, e.g., decomposition of ferrous sulphate on heating: $2\text{FeSO}_4(\text{s}) \xrightarrow{\text{Heat}} \text{Fe}_2\text{O}_3(\text{s}) + \text{SO}_2(\text{g}) + \text{SO}_3(\text{g})$. [2]
9. During respiration, glucose obtained from food combines with oxygen in the cells of the body and breaks down to release energy, along with CO₂ and water as by-products. Since energy (heat) is released during this process, respiration is considered an exothermic reaction.
 $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) + 6\text{O}_2(\text{aq}) \rightarrow 6\text{CO}_2(\text{aq}) + 6\text{H}_2\text{O}(\text{l}) + \text{energy}$ [2]
10. Chemical name: Sodium hydrogencarbonate; Formula: NaHCO₃. Use: It is used as an ingredient in antacids to neutralise excess acid in the stomach and provide relief from acidity (any one valid use, such as use in baking powder, may also be accepted). [2]

SECTION C — Answers

11. A double displacement reaction is a reaction in which two compounds react by an exchange of ions (radicals) between the reactants to form two new compounds, usually with the formation of a precipitate.
 $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$
(1 mark for definition, 2 marks for correctly balanced equation with state symbols) [3]
12. (a) Antacids are mild bases (e.g., magnesium hydroxide) that neutralise the excess hydrochloric acid produced in the stomach during acidity, relieving the pain/irritation, as per the reaction: Base + Acid → Salt + Water.
(b) Toothpaste is basic so that it can neutralise the acids formed in the mouth (due to bacterial breakdown of sugar/food particles), preventing corrosion of tooth enamel and tooth decay, since decay starts when mouth pH falls below 5.5.

(c) Oils and fats in chips undergo oxidation on exposure to air, which makes them rancid and changes their smell and taste. Flushing packets with nitrogen (an inert gas) prevents this oxidation and keeps the chips fresh for longer. [3]

13. Compound A is a metal carbonate (calcium carbonate, CaCO_3), since it reacts with dilute HCl to give brisk effervescence of a gas that extinguishes a flame and turns lime water milky — both are characteristic tests for CO_2 gas.



(1 mark for identifying it as a metal carbonate, 2 marks for the correctly balanced equation) [3]

SECTION D — Answer

14. (a) The chlor-alkali process is the process in which electricity is passed through an aqueous solution of sodium chloride (brine), decomposing it to form sodium hydroxide, chlorine gas and hydrogen gas. It is called 'chlor-alkali' because of the products formed — 'chlor' for chlorine and 'alkali' for sodium hydroxide.



Chlorine gas is liberated at the anode and hydrogen gas at the cathode, while sodium hydroxide solution forms near the cathode. [3 marks]

(b) Uses of the three products:

- Chlorine (Cl_2): used in the manufacture of bleaching powder, and in water treatment/disinfectants, PVC, pesticides, etc. (any one use)
- Hydrogen (H_2): used as a fuel, and in the manufacture of margarine, ammonia for fertilisers, etc. (any one use)
- Sodium hydroxide (NaOH): used in the manufacture of soaps and detergents, de-greasing metals, paper making, artificial fibres, etc. (any one use)

[3 marks — 1 mark for each correct use] [6]